

WAKKANAI, Japan

WMO: 474010

Lat: 45.4153N

Lon: 141.6786E

Elev: 12

StdP: 101.18

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.0	-9.8	-16.7	0.9	-9.3	-15.4	1.0	-8.3	16.2	-1.3	14.3	-1.4	4.2	270	0.631

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.4	24.8	21.6	23.7	20.8	22.8	20.2	22.4	24.1	21.5	23.2	20.7	22.3	5.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.8	16.4	23.7	20.9	15.6	22.8	20.0	14.7	22.0	66.3	24.1	63.1	23.2	60.0	22.3	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
12.4	10.8	9.4	-12.5	26.9	1.7	1.6	-13.7	28.0	-14.7	29.0	-15.7	29.8	-16.9	31.0		
			-13.3	23.0	1.7	1.2	-14.5	23.8	-15.4	24.5	-16.4	25.2	-17.6	26.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.1	-4.6	-4.4	-0.8	4.5	9.3	13.1	17.4	19.8	17.3	11.2	3.7	-2.1
	DBStd	9.03	2.88	3.26	3.10	2.98	3.45	2.98	2.89	2.51	2.59	3.19	4.06	3.14
	HDD10.0	2008	453	404	334	166	56	6	0	0	0	24	191	374
	HDD18.3	4201	711	637	592	414	282	157	53	14	49	221	438	633
	CDD10.0	953	0	0	0	2	33	101	229	304	219	62	3	0
	CDD18.3	104	0	0	0	0	0	1	24	60	19	0	0	0
	CDH23.3	135	0	0	0	0	0	1	19	96	19	0	0	0
	CDH26.7	4	0	0	0	0	0	0	1	2	1	0	0	0
Wind	WSAvg	4.6	5.0	4.7	4.7	4.7	4.7	4.2	3.8	3.9	4.1	4.6	5.0	5.1
Precipitation	PrecAvg	1128	93	65	55	54	72	66	98	113	134	130	127	119
	PrecMax	1755	171	148	140	110	200	162	234	276	330	283	243	192
	PrecMin	779	35	20	10	12	19	3	6	9	30	49	41	35
	PrecStd	200	31	29	25	25	36	37	54	59	68	52	43	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.6	5.1	7.8	15.1	20.4	22.3	25.2	26.6	25.2	19.7	14.7	8.1
		MCWB	1.5	3.4	5.3	9.8	13.5	16.8	21.3	22.7	21.8	16.0	11.9	6.1
	2%	DB	1.7	3.4	6.0	12.3	17.5	20.1	23.6	25.4	23.6	18.4	13.1	5.8
		MCWB	0.0	1.5	3.8	8.3	12.4	16.2	20.2	22.1	20.3	15.1	10.6	3.8
	5%	DB	0.6	2.0	4.8	10.5	15.8	18.7	22.4	24.4	22.4	17.1	11.4	4.0
		MCWB	-0.9	0.0	2.8	7.2	11.7	15.6	19.8	21.6	19.3	13.9	8.7	2.0
	10%	DB	-0.4	0.4	3.7	8.9	14.2	17.4	21.4	23.5	21.3	16.0	9.9	2.4
		MCWB	-1.9	-1.3	1.9	6.2	11.0	14.9	19.4	21.0	18.2	12.7	7.2	0.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.7	3.4	5.6	10.7	14.8	18.3	22.4	23.8	22.5	17.2	12.6	6.6
		MCDB	3.1	4.8	7.4	13.7	17.8	20.8	24.3	25.6	24.4	19.2	14.3	7.8
	2%	WB	0.4	1.8	4.1	8.8	13.4	17.1	21.3	23.0	21.1	15.5	10.8	4.0
		MCDB	1.4	3.2	5.6	11.6	16.4	18.9	22.6	24.6	23.1	17.8	12.9	5.6
	5%	WB	-0.6	0.2	3.1	7.6	12.3	16.1	20.5	22.2	19.8	14.5	9.0	2.2
		MCDB	0.4	1.7	4.6	10.0	15.0	17.9	21.8	23.9	21.8	16.7	11.0	3.8
	10%	WB	-1.9	-1.1	2.1	6.5	11.3	15.3	19.6	21.4	18.7	13.4	7.4	0.6
		MCDB	-0.5	0.3	3.6	8.6	13.7	17.1	21.1	23.1	20.8	15.6	9.6	2.1
Mean Daily Temperature Range	5% DB	MDBR	3.5	4.1	4.2	5.0	5.4	5.0	4.6	4.4	5.0	5.0	4.4	3.7
		MCDBR	4.8	5.5	5.5	7.5	7.7	6.7	5.6	5.3	5.6	5.9	6.4	5.9
		MCWBR	4.6	5.1	4.6	4.5	4.1	3.7	3.3	3.1	3.9	5.0	6.1	5.8
	5% WB	MCDBR	4.6	5.3	5.4	7.0	6.7	5.9	4.9	4.6	5.2	5.7	6.7	5.9
		MCWBR	4.7	5.2	4.9	4.7	4.2	3.7	3.2	3.0	4.1	5.3	6.6	6.0
Clear-Sky Solar Irradiance	taub	0.342	0.389	0.443	0.510	0.528	0.509	0.480	0.451	0.404	0.412	0.391	0.364	
	taud	2.095	1.955	1.889	1.825	1.879	2.024	2.178	2.278	2.343	2.150	2.097	2.034	
	Ebn at Noon	720	750	765	752	755	773	792	802	811	726	653	635	
	Edh at Noon	98	138	170	197	193	168	143	125	107	113	99	95	
All-Sky Solar Radiation	RadAvg	1.10	1.98	3.37	4.54	4.86	4.66	4.46	4.34	3.89	2.59	1.34	0.85	
	RadStd	0.10	0.21	0.28	0.41	0.54	0.52	0.56	0.37	0.36	0.17	0.15	0.07	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page